Lecture 30

Approximating Higher Order Equations

We have developed many different techniques to solve 2nd-order and higher order equations either explicitly or by expanding using a power series. Although these techniques allow us to solve a wide swath of equations, many equations have no explicit or even analytic solution and we must then rely on approximation techniques to find solution values.

We have approximated 1st-order IVPs using various techniques including Euler's method, Taylor methods, and the Runge-Kutta method. In this lecture, we will explore these options for higher order IVPs as well as connect these approximating techniques with other analysis such as phase-plane diagrams and critical point behavior of systems.

Taylor Polynomial Approximations

Given an IVP, we can prescribe a center and assuming it is an ordinary point of the equation, we are guaranteed a collection of analytic solutions depending upon the order of the equation. In such scenarios we can find a Taylor polynomial of any degree to approximate the solution near the center.

x_0 ordinary

$$P_n(x) = \sum_{k=1}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$$

$x=a$ near x_0

$$y(a) \approx P_n(a) = \sum_{k=1}^{n} \frac{f^{(k)}(x_0)}{k!}(a-x_0)^k$$

Example: Consider the IVP

$$y'' - 2y' + y = 2e^x, \quad y(0) = 0, \quad y'(0) = 1.$$

Find the Taylor polynomial for y of degree 4 centered at $x = 0$ and use it to approximate $y(1)$.

- $y(0) = 0$
- $y'(0) = 1$

$y'' = 2e^x + 2y' - y$

- $y''(0) = 2e^0 + 2y'(0) - y(0) = 2 + 2((1)) - 0) = 4$

$y''' = (y'')' = (2e^x + 2y' - y)' = 2e^x + 2y'' - y'$

- $y'''(0) = 2e^0 + 2y''(0) - y'(0) = 2 + 2(4) - 1 = 9$

$y^{(4)} = (y''')' = (2e^x + 2y'' - y')' = 2e^x + 2y''' - y''$

- $y^{(4)}(0) = 2e^0 + 2y'''(0) - y''(0) = 2 + 2(9) - 4 = 16$

So $p_4(x) = \sum_{k=1}^{4} \frac{p^{(k)}(0)}{k!}(x-0)^k = 0 + 1 \cdot x + \frac{4}{2}x^2 + \frac{9}{6}x^3 + \frac{16}{24}x^4$

$\Rightarrow p_4(x) = x + 2x^2 + \frac{3}{2}x^3 + \frac{2}{3}x^4$

So $y(1) \approx p_4(1) = 1 + 2 + \frac{3}{2} + \frac{2}{3} = 3 + \frac{9}{6} + \frac{4}{6} = 3 + \frac{13}{6}$

$= \frac{31}{6}$

$\approx 5.1666\ldots$

Example: Find a Taylor polynomial approximation of degree 4 centered at $x = 0$ for the IVP

$$y' - e^x y = 0, \quad y(0) = 1$$

and use it approximate $y(0.5)$.

$\mu(x) = e^{-\int e^x dx} = e^{-e^x}$

$e^{-e^x} y = C$

$1 = Ce^1$

$C = \frac{1}{e}$

$\boxed{y = Ce^{e^x}}$

$y(0.5) = \frac{e^{e^{0.5}}}{e}$

$\boxed{y(0) = 1}$

$y' = e^x y \Rightarrow y'(0) = e^0 y(0) = 1 \cdot 1 = 1$

$\boxed{y'(0) = 1}$

$y'' = (y')' = (e^x y)' = e^x y + e^x y' = e^x(y + y')$

so $y''(0) = e^0(y(0) + y'(0)) = 1(1+1) = 2 \Rightarrow \boxed{y''(0) = 2}$

$y''' = (y'')' = (e^x(y+y'))' = e^x(y+y') + e^x(y'+y'')$

$= e^x(y + 2y' + y'')$

$y'''(0) = e^0(y(0) + 2y'(0) + y''(0)) = 1(1+2+2) = 5$

$\Rightarrow \boxed{y'''(0) = 5}$

$y^{(4)} = (y''')' = (e^x(y+2y'+y''))'$

$= e^x(y+2y'+y'') + e^x(y'+2y''+y''')$

$= e^x(y + 3y' + 3y'' + y''')$

$\Rightarrow y^{(4)}(0) = e^0(y(0) + 3y'(0) + 3y''(0) + y'''(0))$

$= 1(1+3+6+5) = 15$

$\Rightarrow \boxed{y^{(4)}(0) = 15}$

so $p_4(x) = 1 + x + \frac{2}{2}x^2 + \frac{5}{6}x^3 + \frac{15}{24}x^4$

$= 1 + x + x^2 + \frac{5}{6}x^3 + \frac{5}{8}x^4$

$y(0.5) \approx p_4(0.5) = 1 + 0.5 + (0.5)^2 + \frac{5}{6}(0.5)^3 + \frac{5}{8}(0.5)^4$

$1 + 0.5 + 0.25 + 0.104166 + 0.0390625$

$= \boxed{1.8932285}$

$\frac{e^{e^{0.5}}}{e} = 1.9130929\ldots$

Normal Form and Higher Order Euler's Method

During the study of phase-plane diagrams and 2nd-order mass-spring oscillator equations, we have discussed the conversion of the 2nd-order equation

$$my'' + by' + ky = 0$$

$y_1 = y$

into a system of 1st-order equations by making the substitution $v = y'$ and writing

normal form

$$\begin{cases} y' = v \longrightarrow y' = f(t, y, v) \\ v' = -\frac{b}{m}v - \frac{k}{m}y \longrightarrow v' = g(t, y, v) \end{cases}$$

$y_1' = y_2$, $y_2' = -\frac{b}{m}y_2 - \frac{k}{m}y_1$

This idea can be extended to any order differential equation by making an appropriate number of substitutions. We first begin with a definition.

Definition: A system of m 1st-order equations in m unknown functions $y_1, y_2, y_3, \ldots, y_m$

$$\begin{cases} y_1'(t) = f_1(t, y_1, y_2, \ldots, y_m) \\ y_2'(t) = f_2(t, y_1, y_2, \ldots, y_m) \\ \quad\vdots \\ y_m'(t) = f_m(t, y_1, y_2, \ldots, y_m) \end{cases}$$

is said to be in **normal form**.

For a given system in normal form, we can solve an IVP at t_0 given m initial conditions

$$y_1(t_0) = a_1, \quad y_2(t_0) = a_2, \ldots, y_m(t_0) = a_m$$

We can convert any single differential equation

$$y^{(m)} = f(t, y, y', y'', \ldots, y^{(m-1)})$$

into a system in normal form by making the substitutions

$$y_1 = y, \quad y_2 = y', \quad \ldots \quad , \quad y_m = y^{(m-1)}$$

which gives the following m 1st-order equations

$$\begin{cases} y_1'(t) = y'(t) = y_2 \\ y_2'(t) = y''(t) = y_3 \\ \quad\vdots \\ y_{m-1}'(t) = y^{(m-1)}(t) = y_m \\ y_m'(t) = \boxed{y^{(m)} = f(t, y_1, y_2, \ldots, y_m)} \end{cases}$$

$y_i' = y_{i+1}$, $1 \le i \le m-1$

$y \quad y' \quad y^{(m-1)}$

in normal form.

Example: Convert the IVP

$$y''' + 2y'' + 3y' + 4y = \cos(t), \quad \boxed{y(0) = 1, \quad y'(0) = 2, \quad y''(0) = 3}$$

into an IVP system in normal form.

$\boxed{y_1(0) = 1, \; y_2(0) = 2, \; y_3(0) = 3}$

new conditions

We use $\boxed{y_1 = y, \; y_2 = y', \; y_3 = y''}$

and solve for $\boxed{y''' = \underbrace{\cos(t) - 2y'' - 3y' - 4y}_{f(t, y, y', y'') = f(t, y_1, y_2, y_3)}}$

$$\begin{cases} y_1' = y_2 & , \; y_1(0) = 1 \\ y_2' = y_3 & , \; y_2(0) = 2 \\ y_3' = \cos(t) - 2y_3 - 3y_2 - 4y_1 & , \; y_3(0) = 3 \end{cases}$$

y'''

→ technically in normal form.

Recall, given a 1st-order IVP $y' = f(t, y)$, $y(t_0) = y_0$ we can use Euler's method with step size h to approximate values of y near t_0 with the recursive formulas

$$\begin{cases} t_{n+1} = t_n + h \\ y_{n+1} = y_n + hf(t_n, y_n) \end{cases}$$

where $n + 1$ is the number of steps taken.

$y_1 = y$ solution

Given a system in normal form, we can apply Euler's method to each individual 1st-order equation to approximate values for each function $y_1, y_2, \ldots, y_m$ which corresponds to values of the solution and its derivatives if we have converted an equation into normal form. This is called the **vectorized Euler's method**.

Using step size h the vectorized Euler's method can be written with recursive formulas as

$$\begin{cases} t_{n+1} = t_n + h & \leftarrow \text{step} \\ y_{1:n+1} = y_{1:n} + hf_1(t_n, y_{1:n}, y_{2:n}, \ldots, y_{m:n}) \\ y_{2:n+1} = y_{2:n} + hf_2(t_n, y_{1:n}, y_{2:n}, \ldots, y_{m:n}) \\ \quad\vdots \\ y_{m:n+1} = y_{m:n} + hf_m(t_n, y_{1:n}, y_{2:n}, \ldots, y_{m:n}) \end{cases}$$

step

$(y_1' = f_1(t, y_1, y_2, \ldots, y_m))$

using $y_{i:n}$ to denote the ith function at the nth step. In practice, this collection of equations is written succinctly using vectors (column matrices) and writing

$$\mathbf{y_{n+1}} = \mathbf{y_n} + h\mathbf{f}(\mathbf{t_n}, \mathbf{y_n})$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \text{ and } \mathbf{f}(\mathbf{t}, \mathbf{y}) = \begin{bmatrix} f_1(t, y_1, y_2, \ldots, y_m) \\ f_2(t, y_1, y_2, \ldots, y_m) \\ \vdots \\ f_m(t, y_1, y_2, \ldots, y_m) \end{bmatrix}$$

this vectorized form gives the same visual feel of the original Euler's Method.

Example: Use the vectorized Euler's method to approximate $y'' - 2y' + y = 2e^t$, $y(0) = 0$, $y'(0) = 1$ at $t = 1$ using step size $h = 0.5$.

$t_0 = 0, \quad h = 0.5 \quad$ (2 steps)

<u>Note</u>: For a two equation system $\begin{cases} x' = f(t, x, y), & x(0) = x_0 \\ y' = g(t, x, y), & y(0) = y_0 \end{cases}$ (normal form)

the vectorized Euler's Method is given by

$$\begin{cases} x_{n+1} = x_n + h f(t_n, x_n, y_n) \\ y_{n+1} = y_n + h g(t_n, x_n, y_n) \end{cases}$$

Rewrite $y'' - 2y' + y = 2e^t \Rightarrow y'' = 2e^t + 2y' - y$ set $v = y'$

$$\begin{cases} y' = v \longrightarrow f(t, y, v), & y(0) = 0 = y_0 \\ v' = 2e^t + 2v - y \rightarrow g(t, y, v), & v(0) = 1 = v_0 \end{cases}$$

$$\begin{cases} y_1 = y_0 + h\overbrace{f(t_0, y_0, v_0)}^{v} = 0 + 0.5(1) = 0.5 \\ v_1 = v_0 + h g(t_0, y_0, v_0) = 1 + 0.5(2e^0 + 2(1) - 0) = 1 + 0.5(4) = 3 \end{cases}$$

$t_1 = t_0 + h = 0.5$

$$\begin{cases} y_2 = y_1 + h f(t_1, y_1, v_1) = 0.5 + 0.5(3) = \boxed{2} \\ v_2 = v_1 + h g(t_1, y_1, v_1) = 3 + 0.5(2e^{0.5} + 2(3) - 0.5) \\ \qquad = 3 + 2.75 + e^{0.5} = 5.75 + e^{0.5} \end{cases}$$

approximation of solution.

$y \approx y_2 = 2$ at $t = 1$

Example: Use the vectorized Euler's method with step size $h = 0.25$ to approximate the point (x, y) along the trajectory corresponding to $t = 1$ for the system

$$\begin{cases} x' = \underbrace{1 - y}_{f(t,x,y)}, & x(0) = 1 = x_0 \\ y' = x + \underbrace{1}_{g(t,x,y)}, & y(0) = 1 = y_0 \end{cases}$$

autonomous

$t_0 = 0$, $h = 0.25$ 4 steps

$$\begin{cases} x_1 = x_0 + hf(t_0, x_0, y_0) = 1 + 0.25(1-1) = 1 \\ y_1 = y_0 + hg(t_0, x_0, y_0) = 1 + 0.25(1+1) = 1.5 \end{cases}$$

$$\begin{cases} x_2 = x_1 + hf(t_1, x_1, y_1) = 1 + 0.25(1-1.5) = 1 + 0.25(-0.5) = 0.875 \\ y_2 = y_1 + hg(t_1, x_1, y_1) = 1.5 + 0.25(1+1) = 2 \end{cases}$$

$$\begin{cases} x_3 = x_2 + hf(t_2, x_2, y_2) = 0.875 + 0.25(1-2) = 0.625 \\ y_3 = y_2 + hg(t_2, x_2, y_2) = 2 + 0.25(0.875+1) = 2.46875 \end{cases}$$

$$\begin{cases} x_4 = x_3 + hf(t_3, x_3, y_3) = 0.625 + 0.25(1-2.46875) = 0.2578125 \\ y_4 = y_3 + hg(t_3, x_3, y_3) = 2.46875 + 0.25(0.625+1) = 2.875 \end{cases}$$

$$(x_0, y_0) = (1,1) \to \underset{x_1,\ y_1}{(1, 1.5)} \to \underset{x_2,\ y_2}{(0.875, 2)} \to \underset{x_3,\ y_3}{(0.625, 2.46875)}$$

$$\to \underset{x_4,\ y_4}{(0.2578125, 2.875)}$$

$$\begin{cases} x' = 1-y \\ y' = x+1 \end{cases}$$ the phase-plane equation is

$$\frac{dy}{dx} = \frac{y'}{x'} = \frac{x+1}{1-y} \Rightarrow \int (1-y)\,dy = \int (1+x)\,dx$$

separable

$$\Rightarrow \quad y - \tfrac{1}{2}y^2 = x + \tfrac{1}{2}x^2 + C$$

multiply by 2 $\quad 2y - y^2 = 2x + x^2 + C$

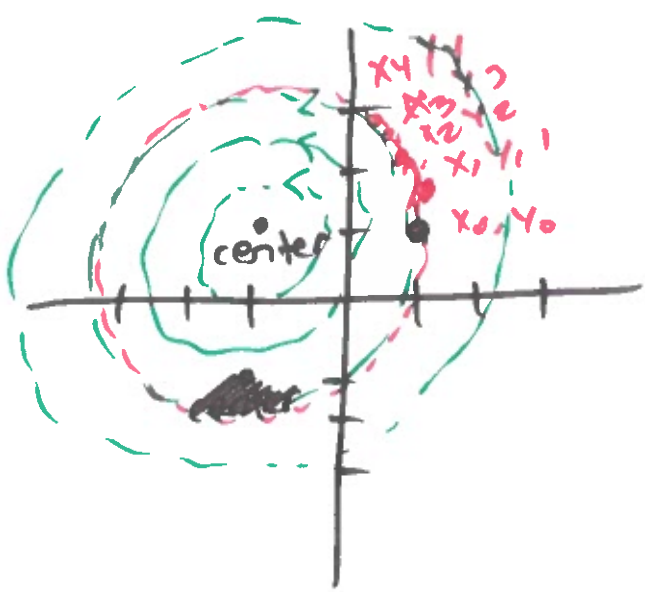
$$\Rightarrow \quad x^2 + 2x + y^2 - 2y = C$$

$$\Rightarrow (x+1)^2 - 1 + (y-1)^2 - 1 = C$$

$$\Rightarrow (x+1)^2 + (y-1)^2 = C$$

circles. center: $(-1,1)$

critical point $1-y=0$, $x+1=0 \Rightarrow x=-1, y=1$



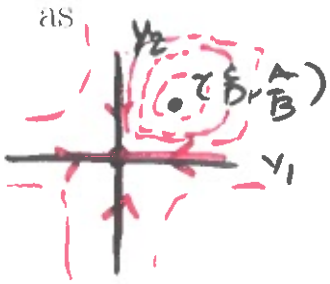
We have discussed various models for population growth and decay such as exponential growth and decay $P' = rP$ and logistic growth $P' = rP(k - P)$. Both of these models discuss the population dynamics of a single population. However, many populations interact in real life and many attempts have been made to construct models detailing the interactions between populations. One such model is a competing species predator-prey model called the **Volterra-Lotka system** developed by A. J. Lotka and V. Volterra. It is represented as an autonomous set of two 1st-order equations

$$\begin{cases} y_1' = Ay_1 - By_1y_2 \\ y_2' = -Cy_2 + Dy_1y_2 \end{cases}$$

normal form

where y_1 is the prey population and y_2 is the predator population. Such a system is not generally explicitly solvable, but we can use our approximating techniques and phase-plane analysis to understand how the particular solutions behave.

Note that the Volterra-Lotka system has two critical lines and a critical point as



$$\begin{cases} y_1' = Ay_1 - By_1y_2 = -By_1\left(y_2 - \dfrac{A}{B}\right) \\ y_2' = -Cy_2 + Dy_1y_2 = Dy_2\left(y_1 - \dfrac{C}{D}\right) \end{cases}$$

$y_1 = 0$ $\quad y_2 = \frac{A}{B}$

$y_2 = 0$ $\quad y_1 = \frac{C}{D}$

which gives $y_1 = 0$, $y_2 = 0$ along with $y_1 = \frac{C}{D}$ and $y_2 = \frac{A}{B}$ as the two critical lines and the critical point. It can be shown that the critical point $(\frac{C}{D}, \frac{A}{B})$ is a *center*. Therefore the trajectories for positive y_1 and y_2 are periodic curves that rotate around the center.

Example: Use the vectorized Euler's method with step size $h = 0.1$ to approximate the solution to

Volterra-Lotka system

$f(t, y_1, y_2) = 2y_1 - 2y_1y_2$

$g(t, y_1, y_2) = y_1y_2 - y_2$

$$\begin{cases} y_1' = 2y_1 - 2y_1y_2, & y_1(0) = 1 \\ y_2' = y_1y_2 - y_2, & y_2(0) = 3 \end{cases}$$

$y_{1;0} = 1$

$y_{2;0} = 3$

at $t = 0.2$

$t_0 = 0, \quad h = 0.1 \quad (1,3) \quad (2 \text{ steps})$

$$\begin{cases} y_{1;1} = y_{1;0} + hf(t_0, y_{1;0}, y_{2;0}) = 1 + 0.1(2(1) - 2(1)(3)) \\ \qquad = 1 + 0.1(2-6) = 1 - 0.4 \\ \qquad = 0.6 \\ y_{2;1} = y_{2;0} + hg(t_0, y_{1;0}, y_{2;0}) = 3 + 0.1(1(3) - 3) = 3 \end{cases}$$

pt $(0.6, 3)$

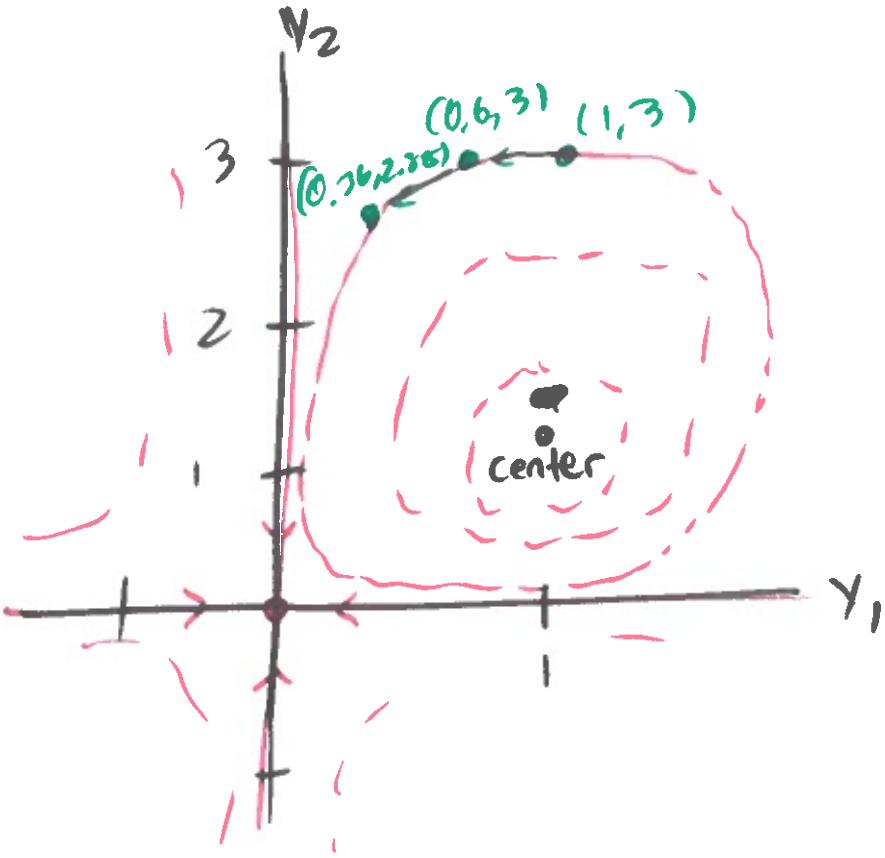
$$\begin{cases} y_{1;2} = y_{1;1} + hf(t_1, y_{1;1}, y_{2;1}) = 0.6 + 0.1(1.2 - 3.6) \\ \qquad = 0.6 - 0.24 \\ \qquad = 0.36 \\ y_{2;2} = y_{2;1} + hg(t_1, y_{1;1}, y_{2;1}) = 3 + 0.1(0.6 \cdot 3 - 3) \\ \qquad = 3 + 0.1(1.8 - 3) \\ \qquad = 3 + (-0.12) = 2.88 \end{cases}$$

$(0.36, 2.88)$

$(1,3) \longrightarrow (0.6, 3) \longrightarrow (0.36, 2.88)$

Here, $A = 2, B = 2, C = 1, D = 1$

critical pt : $\left(\frac{C}{D}, \frac{A}{B}\right) = \left(\frac{1}{1}, \frac{2}{2}\right) = (1,1)$


y_2
3
2
1
(0.6,3)
(1,3)
(0.36,2.88)
center
y_1
1